

David Beckwitt, Ph.D. Candidate

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Physics PhD candidate (July 2026) focused on quantitative X-ray scattering, detector-space simulation, and scientific ML for thin-film analysis.

My work combines X-ray diffraction and reflectivity instrumentation with Python-based simulation, physics-informed machine learning, and AI-assisted scientific-computing workflows to study materials structure and disorder.

Education

- July 2026 📖 **University of Missouri, Columbia, MO — Ph.D. in Physics**
Dissertation: *Investigating Disorder in van der Waals Thin Films*
Advisor: [Dr. Paul Miceli](#)
- May 2022 📖 **University of Missouri, Columbia, MO — M.S. in Physics**
- May 2020 📖 **Missouri State University, Springfield, MO — B.S. in Physics**
Minor: Mathematics, Chemistry

Ongoing Role

- 2021–Present 📖 **Graduate Research Assistant**, University of Missouri
Focus: X-ray diffraction instrumentation, GIWAXS simulation, thin-film growth, physics-informed machine learning, and AI-assisted scientific-computing workflows.
- **Lab operations:** Operate and maintain custom-built in-house x-ray diffraction and reflectivity systems for materials characterization.
 - **Forward modeling:** Built a **Python**-based GIWAXS framework to extract site occupancies, anisotropic Debye-Waller factors, mosaicity, and geometric parameters ([APS Mar 2023](#)).
 - **Defect analysis:** Extended GIWAXS methods to **model diffuse scattering** from stacking faults and **quantify defect densities** ([APS Mar 2024](#)).
 - **Film growth:** Grew phase-controlled PbI₂ films via **Chemical Vapor Deposition** while validating polytype fractions ([ACS AMI 2023](#)).
 - **Scientific ML and AI workflows:** Develop **PyTorch** CNN workflows trained on simulated GIWAXS detector images for structure-analysis screening; use LLM-assisted agents and reusable AI skills for code scaffolding, documentation, reproducibility checks, and validation planning against scattering constraints.

Previous Roles

- 2019–2020 📖 **Research Intern**, NASA Space Consortium
Synthesized graphene films via PLD/PVD; characterized them with Raman spectroscopy and electron microscopy.

Previous Roles (continued)

- 2017–2020  **Research Assistant**, Missouri State University
Advisors: Dr. K. Ghosh and Dr. S. Mitra
Designed and built a PLD system; characterized thin films by XRD, Raman, SEM/EDS, and profilometry ([MRS Spring 2019](#)).
- 2019  **R&D Intern**, Dynatek Labs
Developed software for biomedical testing and automated hardware systems.

Publications

Planned 2026 David Beckwitt et al., *Quantitative X-ray Diffraction Analysis of 2D-Oriented Powders with Area Detectors*.

Selected Presentations

- 2023–2024** Invited talks and presentations on X-ray diffraction at APS Prairie Section 2023, APS March Meeting 2024, ACNS 2024, and Missouri State University Seminar 2024.
- 2020** Fabrication and Characterization of 2D Heterostructures of Graphene and Transition-Metal Oxides.
- 2017–2018** Undergraduate poster presentations on thin-film materials at Einhellig Forum 2017, Einhellig Forum 2018, and Arkansas INBRE 2018.

Technical Skills

-  CORE: **Python** (7+ yrs), C++, Fortran, Git, **MPI**, SQL, LaTeX, Excel/VBA
-  DATA & ML: **NumPy**, pandas, **SciPy**, **PyTorch**, TensorFlow, CNN workflows, scientific image analysis
-  INSTRUMENTATION: **X-ray / neutron scattering**, CVD, PLD, SEM, Raman spectroscopy
-  VISUALIZATION: **Matplotlib**, Plotly, OriginLab, MATLAB, Jupyter, Dash
-  AI-ASSISTED RESEARCH: LLM-assisted agents, reusable AI skills, code scaffolding, literature triage, documentation, reproducibility checks, validation planning; GitHub Copilot, **OpenAI tools**, Claude, **Perplexity**
-  COMMUNICATION: **Technical writing**, grant proposals, peer review, presentations, video editing

Applied Research Tools

- 2025–Present  **2D_Mosaic_Sim** – X-ray Diffraction Simulator
- Visualizes diffraction patterns for materials with specific crystal orientations.
 - Supports analysis beyond standard software through custom plotting and interpretation tools.
- 2024–Present  **ra_sim** – Crystal Analysis Software
- Simulates and analyzes diffraction data from R-Axis IV++ detectors.
 - Combines a focused interface with optimization methods tailored to this detector format.

Applied Research Tools (continued)

2024

📖 **OSC_Reader** – Detector Data Converter

- Converts proprietary detector files into accessible formats.
- Enables detailed inspection of diffraction images without commercial software.

Teaching Experience

2021–Present

📖 **Academic Tutor**, Physics Courses, University of Missouri, Columbia, MO.
Individual and small-group tutoring in undergraduate physics.

2018–2023

📖 **Teaching Assistant**, University of Missouri and Missouri State University **STUDENT REVIEWS**.

Taught and assisted physics courses with enrollments from **15 to 200+ students**.
Selected courses: introductory algebra- and calculus-based mechanics; electricity and magnetism; classical mechanics.

2018–2021

📖 **ACT Prep Tutor**, Club Z!, Springfield, MO.
Individualized test-preparation instruction.

2014–2020

📖 **Martial Arts Coach**, Dunham's Martial Arts, Springfield, MO.
Led skills instruction and student development across ages and experience levels.

Leadership, Service & Outreach

Affiliation	Service	Leadership	Professional Service	Public Engagement
University of Missouri	• CGW Diversity Officer • Graduate Professional Council	• PAGSA President, 2022–2023 • PAGSA Vice President, 2023–2024 • Director, <i>PhysAssist</i>	• Sigma Pi Sigma Physics Congress Judge • X-ray and machine-learning workshops, 2022–present	• MU Extension eclipse events • STEM Cubs • Moberly Correctional Center • Columbia Young Scientist Expo
Missouri State University		• CNAS Leadership Board • Sigma Pi Sigma President		• SPS high school engagement • Physics demonstrations • Student mentoring

Awards

Research	Scholarships / Fellowships	Teaching	Leadership	Travel
• Research & Creative Activity Forum Award, 2025 • NSSA Outstanding Student Research Prize , 2023 • ACNS invited talk, 2023	• Fergason Fund, 2021–2023 • Newell S. Gingrich, 2021–2023 • O.M. Stewart, 2021–2023	• Green Chalk Teaching Award , 2022–2023	• GPC Excellence in Student Leadership, 2022–2023	• Ron Boain & Catherine Rangel-Boain Travel Award, 2023